SAFETY DATA SHEET

April 2024

CEMENTS

- COMMON CEMENT
- MASONRY CEMENT
- BLENDED PORTLAND CEMENTS, BLENDED CEMENTS
- INFRASTRUCTURE PRODUCT RANGE
- HYDRAULIC ROAD BINDERS

In accordance with Regulation (EC) no. 1907/2006 (REACH) and the related appendices

1. IDENTIFICATION OF THE MIXTURE AND THE COMPANY

1.1 Product identifier

Common cements according to EN 197-1 and EN 197-5: CEM I, CEM II, CEM III , CEM V, CEM VI
Blended Portland cement CEM II/C-M and blended cement CEM VI within the meaning of standard NF EN 197-5.
Masonry cements according to EN 413-1: MC
Hydraulic Road Binders according to EN 13282-1

CEM I - Portland Cement: UFI: 5S10-Y05U-900A-XNYN
CEM II/A-S, CEM II/B-S - Slag Portland Cement: UFI: 4V10-F0V7-K00U-M0JS
CEM II/A-V, CEM II/B-V - Fly Ash Portland Cement: UFI: V420-Y0XE-H00A-K1A5
CEM II/A-LL, CEM II/B-LL - Limestone Portland Cement: UFI: E920-00A7-4009-XQGG
CEM II/A-M (LL-S), CEM II/B-M (LL-S) –Portland Composite Cement (limestone, slag): UFI: VD20-H00M-E00T-K22M
CEM II/A-M (LL-P), CEM II/B-M (LL-P) - Portland Composite Cement (limestone, natural pozzolan): UFI: QT20-10FK-Y008-WS08
CEM II/A-P, CEM II/B-P, CEM II/A-Q, CEM II/B-Q – Pozzolan Portland Cement : UFI : J120-G081-600T-XPR1
CEM II/A-M (LL-Q), CEM II/B-M (LL-Q) - Portland Composite Cement (Limestone, calcined natural pozzolan): UFI : QT20-10FK-Y008-WS08
CEM II/C-M (Q-LL) – Portland Composite Cement (Limestone, calcined natural pozzolan): UFI : QT20-10FK-Y008-WS08
CEM III/A, CEM III/B, CEM III/C – Blast furnace cement: UFI: 4V10-F0V7-K00U-M0JS
CEM IV/A (Q), CEM IV/A (P) - Pozzolan Portland Cement: UFI : J120-G081-600T-XPR1
CEM V/A (S-Q) – Composite cement (slag and calcined pozzolan): UFI: W530-2076-6008-J4AU
CEM VI (S-LL) – Composite cement (slag and limestone): UFI: VD20-H00M-E00T-K22M
MC, MC X – Masonry cement: UFI: 1E30-K09D-300R-H537

Hydraulic road binder within the meaning of standard NF EN 13282-1, materials from the Infrastructure range.
The UFI identifiers of the hydraulic road binders and other products in the Infrastructure range are indicated on the delivery documents.

Possible commercial designation:
• Le Classic® • Le Classic® ECOPlanet • L'Original® • Durabat® • Durabat® X-trem • Injektis® • 52.5 R • Superblanc® • Le Liant® • ECOPlanet® • Planet® • Fluid® • Multibat® • Multibat® ECOPlanet • Ingemax® ECOPlanet Prime • CemFrance® • ROLAC® Optimum • ROLAC® Flex • ROLAC® Flex Clair • ROLAC® Performance Immédiate • ROLAC® Premier • Neutracem® • LP01® • Simplicim™
All these products are considered hereafter as 'cement'

1.2 Relevant identified uses of the substance or mixture and uses advised against.

Except the hydraulic road binders, cements are used in industrial installations to manufacture/formulate hydraulic binders for building and construction work, such as ready-mixed concrete, mortars, renders, grouts, plasters as well as precast concrete.
Common cements and cement containing mixtures (hydraulic binders) are used industrially, by professionals as well as by consumers in building and construction work, indoor and outdoor. The identified uses of cements and cement containing mixtures cover the dry products and the products in a wet suspension (paste). See section 16.2 for more information regarding use descriptors and categories.
Cementing of boreholes (cements specified for this use).
Hydraulic road binders: Used for the treatment of soils, embankments, Upper Part of Earthwork (PST), road support platform, capping layer, production of road foundation layers.

Any uses not mentioned above, are advised against. See section 16.2 for the process categories.

1.3 Details of the supplier of the safety data sheet

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Address: 14-16 boulevard Garibaldi, 92130 Issy-les-Moulineaux- France
Phone: +33 (0) 1 58 00 60 00
Fax : +33 (0) 1 58 00 60 02
Email: crc@lafarge.com

1.4 Emergency telephone number

European Emergency call number: 112
Contact Poison Center:

- France : ORFILA (INRS): +33 (0)1 45 42 59 59, http://www.centres-antipoison.net/
- UK: National Poisons Information Service +44 (0) 344 892 0111 (Health professionals only)
- Ireland National Poisons Information Centre : (01) 809 2566
- Switzerland : • Tox Info Suisse ; http://www.toxinfo.ch - Emergency number : 145
- Spain : Poison Information Center: +34 915 620 420.

- Belgium : Poison Information Center : + 32 70 245 245
- Italy :

ROMA CAVp "Osp. Pediatrico Bambino Gesù". +39 06 68593726 - FOGGIA Az. Osp. Univ. Foggia: +39 0881 732326
MILANO Ospedale Niguarda Ca' Granda. Tel: +39 02 66101029 - NAPOLI Ospedale Riuniti Cardarelli. Tel: +39 081 7472870
ROMA Policlinico Agostino Gemelli. Tel: +39 06 3054343 - ROMA Policlinico Umberto I. Tel: +39 06-49978000
PAVIA CAV Centro Naz. di Informazione Tossicologica. Tel: +39 0382 24444 BERGAMO Azienda Ospedaliera Papa Giovanni XXII. Tel.: +39 800 883300
FIRENZE Az. Osp. "Careggi" U.O. Tossicologia Medica. Tel.: 0039 055 7947819 - VERONA Azienda Ospedaliera Integrata. Tel: 800011858

- Sweden : Poisons Information Center:+46 8 33 12 31
- Norway : Poisons Information:+ 47 22 59 13 00
- Denmark : Poison Control Hotline: +45 82 12 12 12
- Germany : Poison Center Information Mainz : +49 6131 19240 ; Berlin : +49 030 30686 790
- Luxembourg : (+352) 8002 5500

2. HAZARDS IDENTIFICATION

2.1 Classification of the substance or mixture

According to Regulation (EC) No 1272/2008 (CLP)

Hazard class	Hazard category	Hazard statements
Skin irritation	2	H315: Causes skin irritation
Serious eye damage/eye irritation	1	H318: Causes serious eye damage
Specific target organ toxicity single exposure respiratory tract irritation	3	H335: May cause respiratory irritation

2.2 Label elements

According to Regulation (EC) No 1272/2008 (CLP)



Danger

Hazard statements

H318 Causes serious eye damage
H315 Causes skin irritation
H335 May cause respiratory irritation

Precautionary statements

P102 Keep out of reach of children
P280 Wear protective gloves/protective clothing/eye protection/face protection
P305+P351+P338+P310: IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Immediately call a POISON CENTER or doctor/physician
P302+P352+P333+P313: IF ON SKIN: Wash with plenty of soap and water. If skin irritation or rash occurs: Get medical advice/attention
P261+P304+P340+P312: Avoid breathing dust/fume/gas/mist/vapours/spray. IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing. Call a POISON CENTER or doctor/physician if you feel unwell.
P501 Dispose of product/packaging by hardening with the application of water and dispose of as concrete waste.

Supplemental information
Skin contact with wet cement, fresh concrete or mortar may cause irritation, dermatitis or burns.
May cause damage to products made of aluminium or other non-noble metals.

2.3 Other hazards

Cement does not meet the criteria for PBT or vPvB in accordance with Annex XIII of REACH (Regulation (EC) No 1907/2006).

Cement dust may cause irritation of the respiratory system.

When cement reacts with water, for instance when making concrete or mortar, or when the cement becomes damp, a strong alkaline solution is produced. Due to the high alkalinity, wet cement may provoke skin and eye irritation.

Cement is either naturally low in soluble chromium VI or reducing agents have been added to control the levels of sensitising soluble chromium (VI) to below 2mg/kg (0.0002%) of the total dry weight of the cement ready for use according to legislation specified under Section 15. If the storage conditions are not appropriate or the storage period is exceeded, the effectiveness of the reducing agent can diminish, and the cement can become skin sensitizing and risk phrase H317 ("May cause an allergic skin reaction") applies to the cement.
In case of atopic dispositions (immediate hypersensitivity type allergy, IgE-dependent) the reactogenic threshold is not subjected to any limit value. Consequently, end users are kindly invited to check their ability to present this atopic disposition and cease any contact in case of immediate reaction. In any case wearing PPI during manipulation is a pre-requisite

3. COMPOSITION/INFORMATION ON INGREDIENTS

3.1 Substances

Not applicable as the product is a mixture, not a substance.

3.2 Mixtures

All type of cements

Substance	Concentration range (m/m) in mixture	REACH Registration number	EINECS	CAS	According to Regulation (EC) No 1272/2008 (CLP)		SCL/ M-factor/ ATE
					Hazard class, category	Hazard statements	
Portland cement clinker	5-100%	NA	266-043-4	65997-15-1	Specific target organ toxicity single exposure respiratory tract irritation 3	H335: May cause respiratory irritation	(Not applicable)
					Skin irritation 2	H315: Causes skin irritation	
					Serious eye damage /eye irritation 1	H318: Causes serious eye damage	
					Skin sensitisation 1B	H317: May cause an allergic skin reaction	
Flue dust from production of cement clinker	0-5%	01-2119486767-17-0003	270-659-9	68475-76-3	Specific target organ toxicity single exposure respiratory tract irritation 3	H335: May cause respiratory irritation	(Not applicable)
					Skin irritation 2	H315: Causes skin irritation	
					Serious eye damage /eye irritation 1	H318: Causes serious eye damage	
					Skin sensitisation 1B	H317: May cause an allergic skin reaction	
Fly Ash	0-35%	01-2119491179-27	268-627-4	68131-74-8	Eye irritation 2	H319: Causes serious eye irritation	(Not applicable)

4. FIRST AID MEASURES

4.1 Description of first aid measures

General notes

No personal protective equipment is needed for first aid responders. First aid workers should avoid contact with wet cement or wet cement containing mixtures

Following contact with eyes

Do not rub eyes in order to avoid possible cornea damage as a result of mechanical stress.
Remove contact lenses if any. Incline head to injured eye, open the eyelid(s) widely and flush eye(s) immediately by thoroughly rinsing with plenty of clean water for at least 20 minutes to remove all particles. Avoid flushing particles into uninjured eye. If possible, use isotonic water (0.9% NaCl). Contact a specialist of occupational medicine or an eye specialist.

Following skin contact

For dry cement, remove and rinse abundantly with water.
For wet cement, wash skin with plenty of water.
Remove contaminated clothing, footwear, watches, etc. and clean thoroughly before re-using them.
Seek medical treatment in all cases of irritation or burns.

Following inhalation

Move the person to fresh air. Dust in the throat and nasal passages should clear spontaneously. Contact a physician if irritation persists or later develops or if discomfort, coughing or other symptoms persist.

Following ingestion

Do not induce vomiting. If the person is conscious, wash out mouth with water and give plenty of water to drink. Get immediate medical attention or contact the anti poison center.

4.2 Most important symptoms and effects, both acute and delayed

Eyes: Eye contact with cement (dry or wet) may cause serious and potentially irreversible injuries.
Skin: Cement may have an irritating effect on moist skin (due to sweat or humidity) after prolonged contact or may cause contact dermatitis after repeated contact.
Prolonged skin contact with wet cement or wet concrete may cause serious burns because they develop without pain being felt (for example when kneeling in wet concrete even when wearing trousers).
For more details see Reference (1)

Inhalation: Repeated inhalation of dust of Common cements over a long period of time increases the risk of developing lung diseases.

Environnement: Under normal use, Common cement is not hazardous to the environment.

4.3 Indication of any immediate medical attention and special treatment needed

When contacting a physician, take this SDS with you.

5. FIRE-FIGHTING MEASURES

5.1 Extinguishing media

Common cements are not flammable

5.2 Special hazards arising from the substance or mixture

Cements are non-combustible and non-explosive and will not facilitate or sustain the combustion of other materials.

5.3 Advice for fire-fighters

Cement poses no fire-related hazards. No need for special protective equipment for fire-fighters.

6. ACCIDENTAL RELEASE MEASURES

6.1 Personal precautions, protective equipment and emergency procedures

6.1.1 For non-emergency personnel

Wear protective equipment as described under Section 8 and follow the advice for safe handling and use given under Section 7.

6.1.2 For emergency responders

Emergency procedures are not required.
However, respiratory protection is needed in situations with high dust levels.

6.2 Environmental precautions

Do not wash cement down sewage and drainage systems or into bodies of water (e.g. streams).

6.3 Methods and material for containment and cleaning up

Collect the spillage in a dry state if possible.

6.3.1 Dry cement

Use cleanup methods such as vacuum clean-up or vacuum extraction (Industrial portable units, equipped with high efficiency air filters (EPA and HEPA filters, EN 1822-1:2009) or equivalent technique) which do not cause airborne dispersion. Never use compressed air.
Alternatively, wipe-up the dust by mopping, wet brushing or by using water sprays or hoses (fine mist to avoid that the dust becomes airborne) and remove slurry.
If not possible, remove by slurrying with water (see wet cement).
When wet cleaning or vacuum cleaning is not possible and only dry cleaning with brushes can be done, ensure that the workers wear the appropriate personal protective equipment and prevent dust from spreading.
Avoid inhalation of cement and contact with skin. Place spilled materials into a container. Solidify before disposal as described under Section 13.

6.3.2 Wet cement

Clean up wet cement and place in a container. Allow material to dry and solidify before disposal as described under Section 13.

6.4 Reference to other sections
See sections 8 and 13 for more details

7. HANDLING AND STORAGE

Do not handle or store near food and beverages or smoking materials.

7.1 Precautions for safe handling
7.1.1 Protectives measures

Follow the recommendations as given under Section 8.
To clean up dry cement, see Subsection 6.3.

Measures to prevent fire
Not applicable.

Measures to prevent aerosol and dust generation
Do not sweep. Use dry cleanup methods such as vacuum clean-up or vacuum extraction, which do not cause airborne dispersion.
For more information, refer to the practice guidelines adopted under the Social Dialogue Agreement on Workers' Health Protection through the Good Handling and Use of Crystalline Silica and Products Containing it, by Employee and Employer European sectoral associations, among which CEMBUREAU. These safe handling practices It can be found via the following link:
https://guide.nepsi.eu/introduction-to-task-manual/

Measure to protect the environment
No particular measures

7.1.1 Information on general occupational hygiene

Do not handle or store near food and beverages or smoking materials.
In dusty environment, wear dust mask and protective goggles.
Use protective gloves to avoid skin contact.

7.2 Conditions for safe storage, including any incompatibilities
Bulk cement should be stored in silos that are waterproof, dry (i.e. with internal condensation minimised), clean and protected from contamination.
Engulfment hazard: To prevent engulfment or suffocation, do not enter a confined space, such as a silo, bin, bulk truck, or other storage container or vessel that stores or contains cement without taking the proper security measures. Cement can build-up or adhere to the walls of a confined space. The cement can release, collapse or fall unexpectedly.
Packed products should be stored in unopened bags clear of the ground in cool, dry conditions and protected from excessive draught in order to avoid degradation of quality.
Bags should be stacked in a stable manner.
Do not use aluminium containers for the storage or transport of wet cement containing mixtures due to incompatibility of the materials.

7.3 Specific end use(s)
No additional information for the specific end uses (see section 1.2).

7.4 Control of soluble Cr (VI)
For cements treated with a Cr (VI) reducing agent according to the regulations given in Section 15, the effectiveness of the reducing agent diminishes with time. Therefore, cement bags and/or delivery documents will contain information on the packaging date, the storage conditions and the storage period appropriate to maintaining the activity of the reducing agent and to keeping the content of soluble chromium VI below 0.0002 % of the total dry weight of the cement ready for use, according to EN 196-10. They will also indicate the appropriate storage conditions for maintaining the effectiveness of the reducing agent.

If shelflife has been exceeded, this does not affect the quality of cement that retains its binding properties. However, its content of soluble chromium VI may exceed 0.0002%, its use will be possible only in controlled closed and totally automated processes excluding contact with skin.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1 Control parameters

8.1.1 DNEL and PNEC

Inhalation DNEL (8hrs): 3 mg/m³
Dermal DNEL: not applicable
Oral DNEL: not applicable

The DNEL value relates to the respirable dust fraction, while the method used for the risk assessment (MEASE) applies to the inhalable fraction. *[Reference (16)]]*. As a result, an additional safety margin is given in the result of the risk assessment and identification of the resulting risk control measures.
There is no known dermal exposure DNEL for workers (either from health hazard studies or human observation). Since cement is an irritant to both the skin and the eyes, it is necessary to minimise skin exposure as much as technically possible.

PNEC water: not applicable
PNEC sediments: not applicable
PNEC soil: not applicable
The environmental risk assessment is based on the impact of pH on water. The final pH of surface water, groundwater and water circulating in treatment plants must not exceed 9.

8.1.2 Occupational exposure limit values:

Europe: Directive (EU) 2017/2398

Substance	CAS Number	Limit of exposure
Crystalline silica, quartz	14808-60-7	TWA: 0.1 mg/m³

France

Type de poussières	Type de valeur limite	VME	Unité	Base légale
Poussières réputées sans effet spécifique	VLEP Poussières totales	4	mg/m³	Article R.4222-10 du Code du Travail
Poussières réputées sans effet spécifique	VLEP Poussières alvéolaires	0,9	mg/m³	Article R.4222-10 du Code du Travail

Case of dust with specific effect:
According to Article R4412-149 of the French Labour Code, the concentration of alveolar quartz dust present in the atmosphere of workplaces must not exceed 0.1 mg/m³ in the workers' breathing zone. For respirable cristobalite and tridymite dusts, the maximum concentration is 0.05 mg/m³. In addition, pursuant to Articles R4412-154 and R4412-155 of the French Labour Code, the concentrations of alveolar quartz, cristobalite and tridymite dust must comply with the following criteria:
Cns/5 + Cq/0.1 + Cc/0.05 + Ct/0.05 ≤ 1
Cns: non-silicogenic respirable dust concentration
Cq,Cc,Ct: concentration of alveolar quartz, cristobalite or tridymite dust.

8.2 Exposure controls

For each individual PROC, users can choose from either option A) or B) in the table above, according to what is best suited to their specific situation. If one option is chosen, then the same option has to be chosen in the table from section "8.2.2 Individual protection measures such as personal protection equipment" - Specification of respiratory protective equipment. Only combinations between A) – A) and B) – B) are possible

8.2.1 Appropriate engineering controls

Measures to reduce generation of dust and to avoid dust propagating in the environment such as dedusting, exhaust ventilation and dry clean-up methods which do not cause airborne dispersion.

Use	PROC*	Exposure	Localised controls	Efficiency
Industrial manufacture/formulation of hydraulic building and construction materials	2, 3	Duration is not restricted (up to 480 minutes per shift, 5 shifts a week)	Not required	-
	14, 26		A) not required or B) generic local exhaust ventilation	- 78 %
	5, 8b, 9		A) general ventilation or B) generic local exhaust ventilation	17 % 78%
Industrial uses of dry hydraulic building and construction materials (indoor, outdoor)	2	Duration is not restricted (up to 480 minutes per shift, 5 shifts a week)	Not required	-
	14, 22, 26		A) not required or B) generic local exhaust ventilation	- 78 %
	5, 8b, 9		A) general ventilation or B) generic local exhaust ventilation	17 % 78%
Industrial uses of wet suspension of hydraulic building and construction materials	7	Duration is not restricted (up to 480 minutes per shift, 5 shifts a week)	A) not required or B) generic local exhaust ventilation	- 78 %
	2, 5, 8b, 9, 10, 13, 14		Not required	-
Professional use of dry hydraulic building and construction material (indoor, outdoor)	2	Duration is not restricted (up to 480 minutes per shift, 5 shifts a week)	Not required	-
	9, 26		A) not required or B) generic local exhaust ventilation	- 72 %
	5, 8a, 8b, 14		A) not required or B) integrated local exhaust ventilation	- 87 %
	19		localised controls are not applicable, process only in good ventilated rooms or outdoor	-
Professional uses of wet suspensions of hydraulic building and construction materials	11	Duration is not restricted (up to 480 minutes per shift, 5 shifts a week)	A) not required or B) generic local exhaust ventilation	- 72 %
	2, 5, 8a, 8b, 9, 10, 13, 14, 19		Not required	-

* PROC's are identified uses and defined in section 16.1

8.2.2 Individual protection measures such as personal protection equipment

General: During work avoid kneeling in fresh mortar or concrete wherever possible. If kneeling is absolutely necessary then appropriate waterproof personal protective equipment must be worn.
Do not eat, drink or smoke when working with cement to avoid contact with skin or mouth.
Before starting to work with cement, apply a barrier cream and reapply it at regular intervals.
Immediately after working with cement or cement-containing materials, workers should wash or shower or use skin moisturisers.
Remove contaminated clothing, footwear, watches, etc. and clean thoroughly before reusing them.

Eye/face protection



Wear approved glasses or safety goggles according to EN 166 when handling dry or wet cement to prevent contact with eyes.

Skin protection



Use watertight, wear- and alkali-resistant protective gloves (eg nitrile soaked cotton gloves with CE marking) internally lined with cotton; boots; closed long-sleeved protective clothing as well as skin care products (eg barrier creams) to protect the skin from prolonged contact with wet cement. Particular care should be taken to ensure that wet cement does not enter the boots. For the gloves, respect the maximum wearing time to avoid skin problems.

In some circumstances, such as when laying concrete or screed, waterproof trousers or knee pads are necessary.

Respiratory protection



When a person is potentially exposed to dust levels above exposure limits, use appropriate respiratory protection. The type of respiratory protection should be adapted to the dust level and conform to the relevant EN standard, (e.g. EN 149, EN 140, EN 14387, EN 1827) or national standard

Thermal hazards

Not applicable

Use	PROC*	Exposure	Specification of respiratory protective equipment (RPE)	RPE efficiency - assigned protection factor (APF)
Industrial manufacture/formulation of hydraulic building and construction materials	2, 3	Duration is not restricted (up to 480 minutes per shift, 5 shifts a week)	not required	-
	14, 26		A) P1 mask (FF, FM) or B) not required	APF = 4 -
	5, 8b, 9		A) P2 mask (FF, FM) or B) P1 mask (FF, FM)	APF = 10 APF = 4
Industrial uses of dry hydraulic building and construction materials (indoor, outdoor)	2	Duration is not restricted (up to 480 minutes per shift, 5 shifts a week)	not required	-
	14, 22, 26		A) P1 mask (FF, FM) or B) not required	APF = 4 -
	5, 8b, 9		A) P2 mask (FF, FM) or B) P1 mask (FF, FM)	APF = 10 APF = 4
Industrial uses of wet suspension of hydraulic building and construction materials	7	Duration is not restricted (up to 480 minutes per shift, 5 shifts a week)	A) P1 mask (FF, FM) or B) not required	APF = 4 -
	2, 5, 8b, 9, 10, 13, 14		not required	-
Professional use of dry hydraulic building and construction material (indoor, outdoor)	2	Duration is not restricted (up to 480 minutes per shift, 5 shifts a week)	P1 mask (FF, FM)	APF = 4
	9, 26		A) P2 mask (FF, FM) or B) P1 mask (FF, FM)	APF = 10 APF = 4
	5, 8a, 8b, 14		A) P3 mask (FF, FM) or B) P1 mask (FF, FM)	APF = 20 APF = 4
	19		P2 mask (FF, FM)	APF = 10
Professional uses of wet suspensions of hydraulic building and construction materials	11	Duration is not restricted (up to 480 minutes per shift, 5 shifts a week)	A) P2 mask (FF, FM) or B) P1 mask (FF, FM)	APF = 10 APF = 4
	2, 5, 8a, 8b, 9, 10, 13, 14, 19		not required	-

* PROC's are identified uses and defined in section 16.1

An overview of the APFs of different RPE (according to EN 529:2005) can be found in the glossary of MEASE (16).
Any RPE as defined above shall only be worn if the following principles are implemented in parallel: The duration of work (compare with "duration of exposure" above) should reflect the additional physiological stress for the worker due to the breathing resistance and mass of the RPE itself, due to the increased thermal stress by enclosing the head. In addition, it shall be considered that the worker's capability of using tools and of communicating are reduced during the wearing of RPE.
For reasons as given above, the worker should therefore be (i) healthy (especially in view of medical problems that may affect the use of RPE), (ii) have suitable facial characteristics reducing leakages between face and mask (in view of scars and facial hair). The recommended devices above which rely on a tight face seal will not provide the required protection unless they fit the contours of the face properly and securely.
The employer and self-employed persons have legal responsibilities for the maintenance and issue of respiratory protective devices and the management of their correct use in the workplace. Therefore, they should define and document a suitable policy for a respiratory protective device programme including training of the workers.

8.2.3 Environmental exposure controls

Environmental exposure control for the emission of cement particles into air has to be in accordance with the available technology and regulations for the emission of general dust particles.
Air: Environmental exposure control for the emission of cement particles into air has to be in accordance with the available technology and regulations for the emission of general dust particles.
Water: Do not wash cement into sewage systems or into bodies of water, to avoid high pH. Above pH 9 negative ecotoxicological impacts are possible.

Soil and terrestrial environment: No special emission control measures are necessary for the exposure to the terrestrial environment.
For further information, refer to Section 6 on 'Accidental release measures'.

9. PHYSICAL AND CHEMICAL PROPERTIES

9.1 Information on basic physical and chemical properties

This information applies to the whole mixture.

(a) Appearance: Dry cement is a finely ground solid inorganic material (grey or white powder). Main particle size: 5-30 µm
(b) Odour: Odourless
(c) Odour threshold: No odour threshold, odourless
(d) pH: (T = 20°C in water, water-solid ratio 1:2): 11-13.5
(e) Melting point: > 1 250 °C
(f) Initial boiling point and boiling range: Not applicable as under normal atmospheric conditions, melting point >1 250°C
(g) Flash point: Not applicable as is not a liquid
(h) Evaporation rate: Not applicable as is not a liquid
(i) Flammability (solid, gas): Not applicable as is a solid which is non combustible and does not cause or contribute to fire through friction
(j) Upper/lower flammability or explosive limits: Not applicable as is not a flammable gas
(k) Vapour pressure: Not applicable as melting point > 1250 °C
(l) Vapour density: Not applicable as melting point > 1250 °C
(m) Relative density: 2.75-3.20; Apparent density: 0.9-1.5 g/cm³
(n) Solubility(ies) in water (T = 20 °C): slight (0.1-1.5 g/l)
(o) Partition coefficient: n-octanol/water: Not applicable as is inorganic mixture
(p) Auto-ignition temperature: Not applicable (no pyrophoricity – no organo-metallic, organo-metalloid or organo-phosphine bindings or of their derivatives, and no other pyrophoric constituent in the composition)
(q) Decomposition temperature: Not applicable as no organic peroxide present
(r) Viscosity: Not applicable as not a liquid
(s) Explosive properties: Not applicable. Not explosive or pyrotechnic. Not in itself capable by chemical reaction of producing gas at such temperature and pressure and at such a speed as to cause damage to the surroundings. Not capable of a self-sustaining exothermic chemical reaction.
(t) Oxidising properties: Not applicable as does not cause or contribute to the combustion of other materials

9.2 Other information

Not applicable.

10. STABILITY AND REACTIVITY

10.1 Reactivity

When mixed with water, cements will harden into a stable mass that is not reactive in normal environments.

10.2 Chemical stability

Dry cements are stable as long as they are properly stored (see Section 7) and compatible with most other building materials. They should be kept dry.

Contact with incompatible materials should be avoided.
Wet cement is alkaline and incompatible with acids, with ammonium salts, with aluminum or other non-noble metals. Cement dissolves in hydrofluoric acid to produce corrosive silicon tetrafluoride gas. Cement reacts with water to form silicates and calcium hydroxide. Silicates in cement react with powerful oxidizers such as fluorine, boron trifluoride, chlorine trifluoride, manganese trifluoride, and oxygen difluoride.

10.3 Possibility of hazardous reactions

Cements do not cause hazardous reactions.

10.4 Conditions to avoid

Humid conditions during storage may cause lump formation and loss of product quality.

10.5 Incompatible materials

Acids, ammonium salts, aluminum or other non-noble metals. Uncontrolled use of aluminum powder in wet cement should be avoided as hydrogen is produced.

10.6 Hazardous decomposition products

Cements will not decompose into any hazardous products.

11. TOXICOLOGICAL INFORMATION

11.1 Information on toxicological effects

Hazard class	C	Effect	Reference
Acute toxicity - dermal	-	Limit test, rabbit, 24 hours contact, 2,000 mg/kg body weight – no lethality. Based on available data, the classification criteria are not met.	(2)
Acute toxicity- inhalation	-	No acute toxicity by inhalation observed. Based on available data, the classification criteria are not met.	(9)
Acute toxicity - oral	-	No indication of oral toxicity from studies with cement kiln dust. Based on available data, the classification criteria are not met.	Literature survey
Skin corrosion/ irritation	2	Cement in contact with wet skin may cause thickening, cracking or fissuring of the skin. Prolonged contact in combination with abrasion may cause severe burns. Some individuals may develop eczema upon exposure to wet cement dust caused by the high pH which induces irritant contact dermatitis after prolonged contact.	(2) Human experience
Serious eye damage/irritation	1	Portland cement clinker caused a mixed picture of corneal effects and the calculated irritation index was 128. Common cements contain varying quantities of Portland cement clinker, fly ash, blast furnace slag, gypsum, natural pozzolans, burnt shale, silica fume and limestone. Direct contact with cement may cause corneal damage by mechanical stress, immediate or delayed irritation or inflammation. Direct contact by larger amounts of dry cement or splashes of wet cement may cause effects ranging from moderate eye irritation (e.g. conjunctivitis or blepharitis) to chemical burns and blindness.	(10), (11)
Skin sensitisation	1B	Some individuals may develop eczema upon exposure to wet cement dust, caused by an immunological reaction to soluble Cr (VI) which elicits allergic contact dermatitis. The response may appear in a variety of forms ranging from a mild rash to severe dermatitis. If the cement contains a soluble Cr (VI) reducing agent and as long as the mentioned period of effectiveness of the chromate reduction is not exceeded, an allergic sensitising effect is not expected [Reference (3)] and a labeling with H317 is not necessary [Reference 18].	(3), (4), (17), (18)
Respiratory sensitisation	-	There is no indication of sensitisation of the respiratory system. Based on available data, the classification criteria are not met.	(1)
Germ cell mutagenicity	-	No indication. Based on available data, the classification criteria are not met.	(12), (13)
Carcinogenicity	-	No causal association has been established between Portland cement exposure and cancer. The epidemiological literature does not support the designation of Portland cement as a suspected human carcinogen Portland cement is not classifiable as a human carcinogen (According to ACGIH A4: Agents that cause concern that they could be carcinogenic for humans but which cannot be assessed conclusively because of a lack of data. In vitro or animal studies do not provide indications of carcinogenicity that are sufficient to classify the agent with one of the other notations.). Based on available data, the classification criteria are not met.	(1) (14)
Reproductive toxicity	-	Based on available data, the classification criteria are not met.	No evidence from human experience
STOT-single exposure	3	Cement dust may irritate the throat and respiratory tract. Coughing, sneezing, and shortness of breath may occur following exposures in excess of occupational exposure limits. Overall, the pattern of evidence clearly indicates that occupational exposure to cement dust has produced deficits in respiratory function. However, evidence available at the present time is insufficient to establish with any confidence the dose-response relationship for these effects.	(1)
STOT-repeated exposure	-	There is an indication of COPD. The effects are acute and due to high exposures. No chronic effects or effects at low concentration have been observed. Based on available data, the classification criteria are not met.	(15)
Aspiration hazard	-	Not applicable as cements are not used as an aerosol.	

Apart from skin sensitisation, Portland cement clinker and Common cements have the same toxicological and eco-toxicological properties.

Medical conditions aggravated by exposure:

Inhaling cement dust may aggravate existing respiratory system disease(s) and/or medical conditions such as emphysema or asthma and/or existing skin and/or eye conditions.

11.2 Information on other hazards

11.2.1 Endocrine disrupting properties

Not applicable.

11.2.2 Other information

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12. ECOLOGICAL INFORMATION

12.1 Toxicity
The product is not hazardous to the environment. Ecotoxicological tests with Portland cement on Daphnia magna [Reference (5)] and Selenastrum coli [Reference (6)] have shown little toxicological impact. Therefore LC50 and EC50 values could not be determined [Reference (7)]. There is no indication of sediment phase toxicity [Reference (8)]. The addition of large amounts of cement to water may, however, cause a rise in pH and may, therefore, be toxic to aquatic life under certain circumstances.

12.2 Persistence and degradability
Not relevant. After hardening, cement presents no toxicity risks.

12.3 Bioaccumulative potential
Not relevant. After hardening, cement presents no toxicity risks.

12.4 Mobility in soil
Not relevant. After hardening, cement presents no toxicity risks.

12.5 Results of PBT and vPvB assessment
Not relevant. After hardening, cement presents no toxicity risks.

12.6 Endocrine disrupting properties
Not relevant.

12.7 Other adverse effects
Not relevant.

13. DISPOSAL CONSIDERATIONS

13.1 Waste treatment methods
Do not dispose of into sewage systems or surface waters.

Product - cement that has exceeded its shelf life
EWC entry: 10 13 99 (wastes not otherwise specified)
(and when demonstrated that it contains more than 0.0002% soluble Cr (VI)): shall not be used/sold other than for use in controlled closed and totally automated processes or should be recycled or disposed of according to local legislation or treated again with a reducing agent.

Product - unused residue or dry spillage
EWC entry: 10 13 06 (Other particulates and dust)Pick up dry unused residue or dry spillage as is. Mark the containers. Possibly reuse depending upon shelf life considerations and the requirement to avoid dust exposure. In case of disposal, harden with water and dispose according to "Product – after addition of water, hardened"

Product – slurries
Allow to harden, avoid entry in sewage and drainage systems or into bodies of water (e.g. streams) and dispose of as explained below under "Product - after addition of water, hardened".

Product - after addition of water, hardened
Dispose of according to the local legislation. Avoid entry into the sewage water system. Dispose of the hardened product as concrete waste. Due to the inertisation, concrete waste is not a dangerous waste.
EWC entries: 10 13 14 (waste from manufacturing of cement – waste concrete or concrete sludge) or 17 01 01 (construction and demolition wastes - concrete).

Packaging
Completely empty the packaging and process it according to local legislation.
EWC entry: 15 01 01 (waste paper and cardboard packaging).

14. TRANSPORT INFORMATION
Cement is not covered by the international regulation on the transport of dangerous goods (IMDG, IATA, ADR/RID), therefore no classification is required.
No special precautions are needed apart from those mentioned under Section 8.

14.1 UN number
Not relevant.

14.2 Nom UN proper shipping name
Not relevant.

14.3 Transport hazard class(es)
Not relevant.

14.4 Packing group
Not relevant.

14.5 Environmental hazards
Not relevant..

14.6 Special precautions for user

Not relevant.

14.7 Transport in bulk according to Annex II of MARPOL73/78 and the IBC Code
Not relevant.

15. REGULATORY INFORMATION

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture
Cement is a mixture according to REACH and is not subject to registration. Cement clinker is exempt from registration (Art 2.7 (b) and Annex V.10 of REACH).

Cement does not meet the criteria for PBT or vPvB in accordance with Annex XIII of REACH (Regulation (EC) No 1907/2006).

The manufacturer declares that it does not include any POP substance (Persistent Organic Pollutants (POP) within the meaning of European Regulation No. 2019/1021 in the cement it produces).

The marketing and use of cement is subject to a restriction on the content of soluble Cr (VI) (REACH Annex XVII point 47 Chromium VI compounds).

National regulatory information (UK):
- CONIAC Health Hazard Information Sheet No 26 (CEMENT)
- Health & Safety at Work, etc. Act 1974
- Control of Substances Hazardous to Health Regulations (COSHH) 2002
- Control of Substances Hazardous to Health (Amendment) Regulations 2004
- Environmental Protection Act 1990
- HSE Guidance Note EH40 (Workplace Exposure Limits)
- Any authorised manual on First Aid by St. John's/St. Andrews/Red Cross Manual Handling Operations Regulations 1992 (as amended)
- PORTLAND CEMENT DUST – criteria document for an occupational exposure limit. June 1994 (ISBN 07176 – 0763 – 1)
- HSE Guidance Notes EH26 (Occupational Skin Diseases – Health and Safety Precautions)

15.2 Chemical Safety Assessment
No chemical safety assessment has been carried out for this mixture by the supplier.

16. OTHER INFORMATION

16.1 Indication of changes

- July 2013: According to EC 1907/2006 + Annex I EC 453/2010
- July 2015: According to EC 1907/2006 + Annex II EC 453/2010
- September 2020: Introducing Hydraulic Road Binders
- December 2021: Introducing Supersulfated Cement acc. EN 15743 and cements acc. EN 197-5.
- May 2022 : Suppression of H317 phrase (according to CEMBureau experts guidelines and (18); addition of the UFI numbers.
- Jan 2023: Updating according to the new Appendix II regulation modified by the (EU) 2020/878 regulation.
- April 2024 :

Section 1.1: Addition of cements containing natural or calcined pozzolan, addition of the brands Ingemax®, SimplicimTM.
Sections 8.1.2 and 15.1: Update of regulatory information.

16.2 Identified uses and use descriptors and categories

The table below gives an overview of all relevant identified uses of cement or cement containing hydraulic binders. All the uses have been grouped in these identified uses because of the specific conditions of exposure for human health and environment. For each specific use, a set of risk management measures or localised controls has been derived (see section 8) which need to be put in place by the user of cement or cement containing hydraulic binders to bring the exposure to an acceptable level

PROC	Identified Uses - Use Description	Manufacture/ Formulation of building and construction materials	Professional/ Industrial use of
2	Use in closed, continuous process with occasional controlled exposure, eg industrial or professional manufacture of hydraulic binders	X	X
3	Use in closed batch process, eg industrial or professional manufacture of ready-mix concrete	X	X
5	Mixing or blending in batch process for formulation of mixtures and articles, eg industrial or professional manufacture of pre-cast concrete	X	X
7	Industrial spraying, eg industrial use of wet suspensions of hydraulic binders by spraying		X
8a	Transfer of substance or mixture from/to vessels/large containers at non-dedicated facilities, eg use of cement in bags to prepare mortar		
8b	Transfer of substance or mixture from/to vessels/large containers a dedicated facilities, eg filling of silos, trucks or barges at cement plants	X	X
9	Transfer of substance or mixture into small containers, eg filling of cement bags in cement plants	X	X
10	Roller application or brushing, eg products to improve adherence between building surfaces and finishing products		X
11	Non-Industrial spraying, eg professional use of wet suspensions of hydraulic binders by spraying		X
13	Treatment of articles by dipping and pouring, eg covering of construction products with a layer to improve the performance of the product		X
14	Production of mixtures or articles by tabletting, compression extrusion, pelletisation, eg production of floor tiling	X	X
19	Hand-mixing with intimate contact and only PPE available, eg mixture of wet hydraulic binder on a construction site		X
22	Potentially closed processing operations with minerals/metals at elevated temperature in industrial setting, eg production of bricks		X
26	Handling of solid inorganic substances at ambient temperature, eg mixture of wet hydraulic binders	X	X

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16.3 Abbreviations and acronyms

ACGIH	American Conference of Industrial Hygienists
ADR/RID	European Agreements on the transport of Dangerous goods by Road/Railway
APF	Assigned protection factor
ATE	Acute toxicity estimate
BOELV	Binding Occupational Exposure Limit Value
CAS	Chemical Abstracts Service
CLP	Classification, labelling and packaging (Regulation (EC) No 1272/2008)
COPD	Chronic Obstructive Pulmonary Disease
DNEL	Derived no-effect level
EC50	Half maximal effective concentration
ECHA	European Chemicals Agency
EINECS	European INventory of Existing Commercial chemical Substances
EPA	Type of high efficiency air filter
ES	Exposure scenario
EWC	European Waste Catalogue
FF P	Filtering facepiece against particles (disposable)
FM P	Filtering mask against particles with filter cartridge
GefStoffV	Gefahrstoffverordnung
HEPA	Type of high efficiency air filter
H&S	Health and Safety
IATA	International Air Transport Association
IMDG	International agreement on the Maritime transport of Dangerous Goods
LC50	Median lethal dose
M-Factor:	Multiplying factor
MEASE	Metals estimation and assessment of substance exposure, EBRC Consulting GmbH for Eurometaux, http://www.ebrc.de/ebrc/ebrc-mease.php
MS	Member State
OELV	Occupational exposure limit value
PBT	Persistent, bio-accumulative and toxic
PNEC	Predicted no-effect concentration
POP	Polluant Organiques Persistants (Persistent Organic Pollutants)
PROC	Process category
RE	Repeated exposure
REACH	Registration, Evaluation and Authorisation of Chemicals
RPE	Respiratory protective equipment
SCOEL	Scientific Committee on Occupational Exposure Limit Values
SCL	Specific concentration limits
SDS	Safety Data Sheet
SE	Single exposure
STP	Sewage treatment plant
STOT	Specific Target Organ Toxicity
SVHC	Substance of Very High Concern
TLV-TWA	Threshold Limit Value-Time-Weighted Average
TRGS	Technische Regeln für Gefahrstoffe
UFI	Unique Formula Identifier
VLE-MP	Exposure limit value-weighted average in mg by cubic meter of air
vPvB	Very persistent, very bio-accumulative
w/w	Weight by weight
WWTP	Waste water treatment plant

16.4 Key literature references and sources of data

(1) *Portland Cement Dust - Hazard assessment document EH75/7*, UK Health and Safety Executive, 2006. Available from: http://www.hse.gov.uk/pubns/web/portlandcement.pdf

(2) *Observations on the effects of skin irritation caused by cement*, Kietzman et al, Dermatosen, 47, 5, 184-189 (1999).

(3) *European Commission's Scientific Committee on Toxicology, Ecotoxicology and the Environment (SCTEE) opinion of the risks to health from Cr (VI) in cement* (European Commission, 2002). http://ec.europa.eu/health/archive/ph_risk/committees/sct/documents/out158_en.pdf

(4) *Epidemiological assessment of the occurrence of allergic dermatitis in workers in the construction industry related to the content of Cr (VI) in cement*, NIOH, Page 11, 2003.

(5) *U.S. EPA, Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms*, 3rd ed. EPA/600/7-91/002, Environmental Monitoring and Support Laboratory, U.S. EPA, Cincinnati, OH (1994a) and 4th ed. EPA-821-R-02-013, US EPA, office of water, Washington D.C. (2002).

(6) *U.S. EPA, Methods for Measuring the Acute Toxicity of Effluents and Receiving Waters to Freshwater and Marine Organisms*, 4th ed. EPA/600/4-90/027F, Environmental Monitoring and Support Laboratory, U.S. EPA, Cincinnati, OH (1993) and 5th ed. EPA-821-R-02-012, US EPA, office of water, Washington D.C. (2002).

(7) *Environmental Impact of Construction and Repair Materials on Surface and Ground Waters. Summary of Methodology, Laboratory Results, and Model Development*. NCHRP report 448, National Academy Press, Washington, D.C., 2001.

(8) *Final report Sediment Phase Toxicity Test Results with Corophium volutator for Portland clinker* prepared for Norcem A.S. by AnalyCen Ecotox AS, 2007.

(9) TNO report V8801/02, *An acute (4-hour) inhalation toxicity study with Portland Cement Clinker CLP/GHS 03-2010-fine in rats, August 2010.*

(10) TNO report V8815/09, *Evaluation of eye irritation potential of cement clinker G in vitro using the isolated chicken eye test*, April 2010

(11) TNO report V8815/10, *Evaluation of eye irritation potential of cement clinker W in vitro using the isolated chicken eye test*, April 2010

(12) *Investigation of the cytotoxic and proinflammatory effects of cement dusts in rat alveolar macrophages*, Van Berlo et al, Chem. Res. Toxicol., 2009 Sept; 22(9):1548-58.

(13) *Cytotoxicity and genotoxicity of cement dusts in A549 human epithelial lung cells in vitro*; Gminski et al, Abstract DGPT conference Mainz, 2008

(14) *Comments on a recommendation from the American Conference of governmental industrial Hygienists to change the threshold limit value for Portland cement*, Patrick A. Hessel and John F. Gamble, EpiLung Consulting, June 2008

(15) *Prospective monitoring of exposure and lung function among cement workers, Interim report of the study after the data collection of Phase I-II 2006-2010*, Hilde Notø, Helge Kjuus, Marit Skogstad and Karl-Christian Nordby, National Institute of Occupational Health, Oslo, Norway, March 2010,

(16) *MEASE, Metals estimation and assessment of substance exposure,* EBRC Consulting GmbH for Eurometaux, http://www.ebrc.de/ebrc/ebrc-mease.php

(17) *Occurrence of allergic contact dermatitis caused by chromium in cement. A review of epidemiological investigations. Kåre Lenvik, Helge Kjuus, NIOH, Oslo, December 2011.*

(18) ECHA Support Questions and answers agreed with National Helpdesks. ID1695 May 2020. https://echa.europa.eu/es/support/qas-support/qas-agreed-with-national-helpdesks

16.5 Current Hazard statements and precautionary statements

Hazard statements and precautionary statements are already listed under Section 2 'Hazards identification', 2.1 'Classification of the substance or mixture' and 2.2 Label elements'.

16.6 Training advice

In addition to health, safety and environmental training programs for their workers, companies must ensure that workers read, understand and apply the requirements of this SDS.

16.7 Further information

See Annex(es) for the ES of the following substances:
The data and test methods used for the purpose of classification of Common cements are given or referred to in section 11.1.

16.8 **Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]**

Classification according to Regulation (EC) No. 1272/2008	**Classification procedure**
Skin Irrit. 2, H315	on basis of test data
Eye dam. 1, H318	on basis of test data
Skin sens. 1B, H317	Human experience
STOT SE. 3, H335	Human experience

16.9 Disclaimer

The information on this data sheet reflects the currently available knowledge and is reliable provided that the product is used under the prescribed conditions and in accordance with the application specified on the packaging and/or in the technical guidance literature. Any other use of the product, including the use of the product in combination with any other product or any other process, is the responsibility of the user.
It is implicit that the user is responsible for determining appropriate safety measures and for applying the legislation covering his/her own activities.